

DAY 02 · LAB

Trace the Route.

DURATION

90 minutes

STRUCTURE

17 steps, 5 parts

WORKING

Alone, then teams of five

ACCOUNTS

None

Interrogate the naming system. Follow the packets.
Then **negotiate** for the right to carry them.

You will find out who really serves a website, follow packets to five countries, identify every machine that touched them, check the answers against the speed of light, and draw the result on a map.

Then you put the laptops down and argue about money, because that is what actually decides how your traffic gets anywhere.

PART	STEPS	TIME
Tools, and where you are	1 to 2	5 min
Unmask the CDN	3 to 6	15 min
One trace, together	7 to 9	12 min
Five countries	10 to 11	18 min
Ownership and physics	12 to 14	15 min
The Peering Table	15 to 16	17 min
Findings	17	8 min

5 MINUTES

01

Confirm your tools work

Run one command for your platform. You are checking that something happens, not reading the output yet.

```
macOS / Linux (Terminal)
tracert -q 1 -w 2 -m 30 www.mit.edu
dig www.mit.edu

Windows (PowerShell)
tracert -w 1000 -h 30 www.mit.edu
Resolve-DnsName www.mit.edu
```

If nothing works, or you have no terminal. Everything in this lab runs in a browser instead. You will be slower and learn the same things. Use **globalping.io** for traces, **whatsmydns.net** for names.

02

Find out who you are

Open `https://ipinfo.io/json` and record what it says.

My public IP

My ASN

Operator name

City it thinks I am in

That AS number is your network's identity in the global routing table. There are roughly **80,000** of them. Two caveats: the address shown is not the one on your laptop, and the city is a commercial guess that is sometimes wrong by thousands of kilometres.

————

15 MINUTES · BEFORE YOU FOLLOW A SINGLE PACKET, FIND OUT WHO YOU ARE TALKING TO

03

Turn a name into an address

Pick a large commercial website. A news organisation, an airline, a major retailer.

```
macOS / Linux
dig www.example.com +short

Windows PowerShell
Resolve-DnsName www.example.com

Any browser
whatsmydns.net, record type A
```

You will often get **several addresses for one name**. That is the first clue. A single machine has one address. A name that answers with four is not a machine, it is a system.

04

Ask three resolvers the same question

This is the step that gives the game away.

```
macOS / Linux
dig @1.1.1.1 www.example.com +short
dig @8.8.8.8 www.example.com +short
dig @9.9.9.9 www.example.com +short

Windows PowerShell
Resolve-DnsName www.example.com -Server 1.1.1.1
Resolve-DnsName www.example.com -Server 8.8.8.8
Resolve-DnsName www.example.com -Server 9.9.9.9
```

1.1.1.1 says

.....

8.8.8.8 says

.....

9.9.9.9 says

.....

Same answer?

.....

Y / N

WHAT YOU JUST LEARNED

For a small site all three agree. For a large one they differ, and differ again when asked from another country. **The name does not identify a machine. It identifies whichever machine is nearest to whoever is asking.** The naming system is doing geography before a single packet has been sent.

05

Follow the chain, name the company

Ask again without `+short` and read the whole answer. In the ANSWER section look for **CNAME** lines, which are aliases. One name points at another, and the last one in the chain is the company actually serving the site.

```
macOS / Linux      Windows PowerShell
dig www.example.com  Resolve-DnsName www.example.com -Type A

www.example.com
→ -----
→ -----
→ -----
```

Test at least four sites. A large commercial one, a government one, a news organisation, and a university.

MATCH THE END OF THE CHAIN

IF THE CHAIN ENDS IN	THE SITE IS SERVED BY
akamaiedge.net, akadns.net, edgekey.net, edgesuite.net	Akamai
cloudfront.net	Amazon
fastly.net, fastlylb.net	Fastly
cloudflare.net	Cloudflare
azureedge.net, trafficmanager.net	Microsoft
linwd.net · edgecastcdn.net	Limelight · Edgecast
Nothing. Just an address	The organisation serves itself

A handful of companies serve an enormous share of the web, and their names never appear in your address bar. You have just read the ownership structure of the internet off the naming system, and none of it was secret.

06

Watch a cache expire

Run the same query three times, about twenty seconds apart, and watch the number in the TTL column.

```
macOS / Linux
dig www.example.com +noall +answer

Windows PowerShell
Resolve-DnsName www.example.com | Select-Object Name,Type,TTL

First run,  TTL: -----
Second run, TTL: -----
Third run,  TTL: -----
```

WHAT THAT NUMBER IS

A countdown, in seconds. It is your resolver telling you how long it will keep reusing this answer before asking again.

When it reaches zero, the next question goes all the way back to the authoritative server.

That single number is why the internet does not collapse under the weight of everybody asking the same question. It is also why a change to a website can take hours to reach everybody.

Checkpoint 1. You should now be able to name the company that serves at least one major website, without that company being mentioned anywhere on the site. If you cannot, find someone who can before moving on.

12 MINUTES · YOUR INSTRUCTOR RUNS THIS ON THE PROJECTOR. FOLLOW ON YOUR OWN MACHINE

07

Run the trace

```
macOS / Linux
tracert -q 1 -w 2 -m 30 www.u-tokyo.ac.jp

Windows PowerShell
tracert -w 1000 -h 30 www.u-tokyo.ac.jp
```

Each line is one hop, meaning one router that handled your packet and passed it on. The number is the round trip to **that router**, not the time from the previous one. So the numbers should climb.

09

Find the ocean

Look down the latency column for the single biggest jump between two consecutive hops.

Between hop ____ and hop ____

From ____ ms to ____ ms

A jump of 60 ms or more in one hop is almost always **water**. You have just located a submarine cable in a list of numbers.

08

Divide the output into three zones

Draw two horizontal lines across your output.

ZONE	WHAT IT IS	HOW TO SPOT IT
Your building	Campus or ISP network	Hops 1 to 3, low numbers, addresses like 10.x or 192.168.x
The middle	Transit carriers and exchange points	Airport codes and carrier names, latency climbing
The destination	The target's own network	Last few hops, names matching the organisation

The middle zone belongs to nobody you have a relationship with. You did not choose those companies, you cannot contact them, and your traffic depends on them entirely.

18 MINUTES • ON YOUR OWN NOW

10

Choose and trace five targets

At least four continents. Include at least one site from Step 5 that uses a CDN, and at least three universities, which usually host their own servers. **That contrast is deliberate and it matters in Step 13.**

TARGET	COUNTRY	WHY IT IS ON THE LIST
www.mit.edu	United States	Large, likely behind a CDN
www.u-tokyo.ac.jp	Japan	University, likely self hosted
www.uct.ac.za	South Africa	Long haul, few cable paths
www.unsw.edu.au	Australia	Trans-Pacific or trans-Indian
www.usp.br	Brazil	South Atlantic
www.iitb.ac.in	India	Several plausible paths

11

Record each one

Target and resolved IP

Hops to reach it

Final round trip

Biggest single jump

Number of star lines

Hostnames with a place

Five times. Much of it is typing, which is fine. **Step 13 is where it pays off.**

12

Identify six hops

On your longest trace, work out who owns the router and roughly where it is. Three sources, in order of reliability.

1. The hostname. Carriers name routers in patterns, often containing a three letter airport code.

2. The AS number. Look the address up at ipinfo.io or bgp.he.net. This answer is reliable.

3. The latency. If the round trip is 12 ms, the router is not on another continent, whatever a database claims. **Physics beats databases.**

AIRPORT CODES YOU WILL ACTUALLY SEE

ams	Amsterdam	fra	Frankfurt	nrt	Tokyo	syd	Sydney
atl	Atlanta	gru	São Paulo	ord	Chicago	mel	Melbourne
bom	Mumbai	hkg	Hong Kong	ruh	Riyadh	yyz	Toronto
cdg	Paris	iad	Washington	sin	Singapore	icn	Seoul
dxb	Dubai	jed	Jeddah	sjc	San Jose	mad	Madrid
dub	Dublin	jnb	Johannesburg	lhr	London	lax	Los Angeles

Other patterns. `xe-`, `ae`, `be-` and `100ge` are interface names, not places. `core`, `bb` and `edge` tell you where in the network you are. `pe` and `ce` mean provider edge and customer edge, so you are standing at a boundary between two organisations, and boundaries are where money changes hands.

13

Check it against the speed of light

Light in glass travels at about **200,000 km per second**, two thirds of its speed in a vacuum.

Minimum round trip in ms

= distance in km ÷ 100

200,000 km/s is 200 km per millisecond. A round trip covers twice the distance, so the time is $2d \div 200$, which is $d \div 100$.

Straight line distance	KM
Theoretical minimum	MS
Measured	MS
Reality tax	MEASURED ÷ MIN

REALITY TAX

WHAT IT MEANS

Below 1.0	Impossible. You never reached that country. Something answered locally
1.3 to 2.0	Normal. Cables do not run in straight lines and routers take time
2.0 to 3.0	A detour. Your traffic probably went via a third continent
Above 3.0	A trombone route, congestion, or a satellite link. Investigate

A reality tax below 1.0 is the best result you can get. **Cross-check it against Step 5.** If that site had a chain ending in a CDN's name, you have now proved the same fact twice: once from the naming system, once from physics.

14

Draw it

- 01 Open `geojson.io` or Google My Maps
- 02 Drop a marker for each hop you located, in order
- 03 Join them with a line
- 04 Open `submarinecablemap.com` alongside it
- 05 Name the cable your packet most likely used for the ocean crossing you found in Step 9
- 06 Export or screenshot

CHECKPOINT 2

Before Part 5 you should have:

- Four sites, with the company that really serves each one named
- Five traces recorded
- Six hops identified by owner and location
- At least one reality tax calculated
- One map with a named cable on it

If the map is not finished, leave it. Do it afterwards and hand it in with the deliverable.
Do not miss Part 5.

17 MINUTES · LAPTOPS DOWN. GROUPS OF FIVE. ONE ROLE CARD EACH, SHOWN TO NOBODY

15

The rules

You are five networks. Every network must be able to reach every other network, or your customers cannot use the internet. There are exactly two ways to achieve that.

OPTION	COST	CONDITIONS
Peer	5 credits each	A direct link to one network. Both sides must agree. Reaches that network only
Buy transit	30 credits	A carrier reaches everybody on your behalf. Always available, nobody's permission needed

THE ARITHMETIC

With four other networks, peering with all of them costs **20**. Transit costs **30**.

So peering is always cheaper, if everybody agrees.

They will not all agree.

Ten minutes to negotiate. Talk to anyone, in any order. Write down every deal and what it cost. At the end you must be able to reach all four other networks, one way or another.

16

Reveal, and put the totals on the board

Go around the table. Who did you peer with, who did you pay, and what did it cost you in total?

NETWORK	PEERED WITH	PAID TRANSIT?	TOTAL SPEND
Streaming service			
National ISP			
Transit carrier			
University network			
Remote ISP			

THE QUESTION TO SIT WITH

The streaming service sent more traffic than everybody else combined.

Check the board. Did they pay the most?

Whatever your table produced, one thing will hold: the amount you paid had very little to do with how much traffic you sent, and a great deal to do with whether anybody else needed you.

8 MINUTES · IN WRITING, 300 WORDS IS PLENTY

- 01

In Step 5, which company served the most sites you tested? Their name appeared nowhere in your address bar. Why not?
- 02

Which target had the highest reality tax, and what is your best explanation?
- 03

Which trace never left your continent? What does that tell you about where that content is stored, and about who decided to put it there?
- 04

Where did you see stars? Give three possible explanations and say which you believe in this case.
- 05

In the Peering Table, who paid the most, and why? Was it the network that sent the most traffic, or somebody else?
- 06

Looking at your map: if one cable were cut tomorrow, which would hurt you most, and what would your traffic do instead?

THE CABLE CUT WARGAME

On the submarine cable map, pick a country served by exactly two cables. Cut one.

Where does its traffic go? How much further is that? What happens to the surviving cable, which was never sized to carry double?

Then find a country served by **one** cable, and consider what that means for its government's negotiating position on almost anything.

ROUTE ASYMMETRY

Using globalping, traceroute the same target from five different countries.

The paths will not be reverses of each other, and they will not agree.

There is no single "the internet". There is only the internet as seen from where you happen to be standing.

THE EXCHANGE POINT CENSUS

Find the largest internet exchange point in your country on [peeringdb.com](https://www.peeringdb.com).

How many networks are present? Who runs it? Who owns the building?

Then ask the uncomfortable question: what happens to your country's internet if that one building has a fire?

SYMPTOM	CAUSE	FIX
Stars after hop 1	The campus firewall drops the probes	<code>tracert -T -p 443</code> , or globalping.io
The final hop never answers	The destination ignores these packets by design	Normal. The second last hop is your answer
dig not found on Windows	It is a unix tool and is not installed	Use <code>Resolve-DnsName</code> . See the Windows card
Resolve-DnsName not recognised	You are in Command Prompt, not PowerShell	Windows key, type PowerShell, open that
All three resolvers agree	The site is small, or not using a CDN	That is a finding. Try a larger site
Eight hops, identical latency	A tunnel is hiding the middle of a network	Nothing to fix. Note it, it is real
Two runs, different paths	Load balancing across parallel links	Expected. Use <code>mtr</code> or <code>-q 3</code>

HAND IN

Four companies serve most of what you read. Nobody **voted** for that.

ONE

The CDN table from Step 5, four sites

TWO

Five routes, and six hops identified

THREE

One map with a named cable

FOUR

Your Peering Table result and total spend

FIVE

The six answers from Step 17

PRINT THIS FOR THE WINDOWS USERS IN THE ROOM. EVERYTHING IN THIS LAB, BOTH DIALECTS

WHAT YOU WANT	MACOS / LINUX	WINDOWS POWERSHELL
Trace a route	tracert -q 1 -w 2 -m 30 HOST	tracert -w 1000 -h 30 HOST
Trace, with statistics	mtr --report HOST	pathping -q 10 -p 100 HOST
Resolve a name	dig HOST +short	Resolve-DnsName HOST
Ask a specific resolver	dig @1.1.1.1 HOST +short	Resolve-DnsName HOST -Server 1.1.1.1
See the alias chain	dig HOST	Resolve-DnsName HOST -Type A
See the TTL	dig HOST +noall +answer	Resolve-DnsName HOST Select Name,Type,TTL
Find your own address	Open <code>https://ipinfo.io/json</code> in a browser. Same on every platform	
Trace from another country	globalping.io in a browser. Same on every platform	

FIVE TRAPS, AND THEY ARE SILENT

- **grep does not exist.** Nothing in this lab needs it. Read the output, or use `Select-String`
- **nslookup works but hides alias chains,** which is the whole point of Step 5. Use `Resolve-DnsName` instead
- **sudo does not exist.** Right-click PowerShell and choose Run as administrator
- **tracert sends three probes per hop,** so it is slower than traceroute. The `-w 1000` caps the wait per hop
- **Command Prompt is not PowerShell.** `Resolve-DnsName` needs PowerShell. Press the Windows key and type PowerShell

INSTRUCTOR • PRINT AND CUT BEFORE THE SESSION. DO NOT PROJECT THIS PAGE

CARD 1

Global streaming service

You send enormous volumes of traffic and receive almost none. Everybody's customers want what you have.

Budget: 40 credits.

Your position. Peering is worth it to you only with networks that have a lot of subscribers. You have no reason to pay anybody to carry traffic their own customers are demanding.

CARD 2

National ISP, 8 million subscribers

You receive enormous volumes and send little. Your customers complain loudly when video is slow.

Budget: 40 credits.

Your position. You would like the streaming service to pay you for access to your subscribers. They would like the opposite. Somebody has to blink.

CARD 3

Tier-1 transit carrier

You own long distance cables and you sell reachability.

Budget: 60 credits.

Your position. You peer only with networks of comparable size and reach. Everyone else is a customer, not a peer. You make money from networks that cannot peer with each other.

CARD 4

University and research network

Modest traffic both ways, valuable research data, a non-commercial mandate.

Budget: 20 credits.

Your position. You peer freely with other research networks and with anyone who will have you. You have very little money and no commercial leverage.

CARD 5

ISP in a small, remote market

Two million subscribers, a long way from everybody, one submarine cable.

Budget: 25 credits.

Your position. You need to reach everyone. Nobody particularly needs to reach you.